

CS101 – INTRODUCTION TO COMPUTING

Full marks – 132

25th June, 2022

Name : _____

Roll No. :

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Instructions

1. There are 45 questions in all and you are required to attempt as many questions as you can.
2. A separate **answer sheet** has been provided to you for writing your solutions, along with rough sheet.
3. No clarifications will be given during the examination.
4. Some multiple choice questions may have more than one option as the correct answer.
5. No partial marks will be given for any question.

1. Consider the following declaration.

```
int a[10];
```

Which of the following is true? [1]

- (a) **a** is a variable.
- (b) **a** is a constant.
- (c) **a** is a pointer to an integer.
- (d) None of the above.

2. Consider the following piece of code.

```
int a[10];  
int *p;
```

```
p = a;
```

Which of the following expressions refer to the second element of the array? [2]

- (a) `p[1]`
- (b) `*p + 1`
- (c) `*(p + 1)`
- (d) `*(a + 1)`
- (e) None of the above

3. Consider the following piece of code.

```
int a[10];  
int *p;  
  
p = &a[9];
```

Which of the following expressions give the size of the array? [2]

- (a) `p - 1`
- (b) `p - &a[0]`
- (c) `p - a + 1`
- (d) None of the above.

4. Consider the following declaration.

```
int a[10];
```

What is the data type of the address of `a[0]`? [1]

- (a) `int`
- (b) `int *`
- (c) `int **`
- (d) `int ***`

5. Consider the following piece of code.

```

int i, j;

for (i = 0; i < 5; ++i)
{
    for (j = 0; j == i; ++j)
    {
        printf("%d", j);
    }
}

```

What will be the output of the code? [2]

6. Consider the following recursive function.

```

int q(int a, int b)
{
    if (a == b)
    {
        return 1;
    }
    else if ( a % b )
    {
        return q(a, b+1);
    }
    else
    {
        return 0;
    }
}

```

What will the function return when called as q(35, 2) and as q(11, 2)? [5]

7. After reviewing the given code, identify the correct statement or statements from the following list. [2]

```

int a[3] = { 1, 2, 3 };
int *p;
int s, i;

p = &a[0];
s = 0;

for (i = 0; i < 3; ++i)
{
    s = s + *(p + i);
}

printf("%d", s);

```

- (a) The code will produce a compilation error.
- (b) The code will correctly compile but will produce a runtime error.
- (c) The code will print 6.

8. Which of the following expressions evaluate to the base address of the array given below? [3]

```
int a[10];
```

- (a) a
- (b) *a
- (c) &a[0]
- (d) &a[1] - 1

9. Consider the following declarations.

```

struct s
{
    int t;
};

struct s u;
struct s *p = &u;

```

Which of the expressions correctly access the member t of the structure u? [3]

- (a) (*p).t
- (b) (*(&s))->t
- (c) p->t
- (d) s.t

10. Consider the following piece of code.

```

int i, j, k;

for (i = 2; i <= k; ++i)
{
    for (j = 2; j <= i; ++j)
    {
        if (i == j)
        {
            printf("%d", i);
        }
        else if (i%j == 0)
        {
            break;
        }
    }
}

```

State clearly the numbers which will be printed by the code when k is 10. [5]

11. Consider the following declaration.

```
int a[10][20];
```

Which of the following expressions refer to the element a[2][3]? [3]

- (a) *(*(a + 2) + 3)
- (b) *(a[2] + 3)

- (c) $*(a[0] + 2*20 + 3)$
 (d) None of the above.

12. In the declaration

```
int *a[10];
```

what is the data type of the address of the element $a[0]$? [2]

13. Consider a linked list whose individual node has the data type –

```
struct node
{
    int data;
    struct node *ptr;
};
```

The first node's address is stored in the following variable.

```
struct node *start;
```

Which of the following conditions will be true if the linked list has exactly two nodes? [2]

- (a) $start == NULL$
 (b) $start \rightarrow ptr == NULL$
 (c) $start \rightarrow ptr \rightarrow ptr == NULL$
 (d) $start \rightarrow ptr \rightarrow ptr \rightarrow ptr == NULL$

14. Consider the following piece of code, where the condition in the for loop has been omitted.

```
for (p = start;      ; p = p->ptr)
{
    /* Empty for loop; do nothing */
}
```

What could be a suitable condition in the for loop such that when the loop terminates p points to the second last node? You can assume that the linked list has at least two nodes. [3]

15. Consider the following code to remove the first node from a non-empty linked list.

```
if ( A )
{
    start = NULL;
}
else
{
    B
}
```

State an appropriate condition for A and a single statement for B. [4]

16. Consider the following function on linked lists.

```
void fun(struct node* head)
{
    if (head == NULL)
        return;

    fun(head->ptr);

    printf("%d ", head->data);
}
```

What does the following function do to a linked list if it is passed the address of the first node of the linked list? [3]

- (a) Prints all the nodes of the linked list in the order of their occurrence in the list starting from the first node.
 (b) Prints all the nodes of the linked list in reverse order.
 (c) Prints alternate nodes of the linked list.
 (d) Prints alternate nodes of the linked list, but in reverse order.

17. The following C function takes a singly linked list as input argument. It modifies the list by moving the last element to the front of the list and returns the modified list. Some part of the code has been omitted.

```
struct node *move_to_front(struct node *head)
{
    struct node *p, *q;

    if((head == NULL) || (head->ptr == NULL))
        return head;

    q = NULL; p = head;

    while (p->ptr != NULL)
    {
        q = p;
        p = p->ptr;
    }

    -- A --

    return head;
}
```

Choose the correct alternative to replace the letter A. [5]

- (a) $q = NULL; p \rightarrow ptr = head; head = p;$
 (b) $q \rightarrow ptr = NULL; head = p; p \rightarrow ptr = head;$
 (c) $head = p; p \rightarrow ptr = q; q \rightarrow ptr = NULL;$
 (d) $q \rightarrow ptr = NULL; p \rightarrow ptr = head; head = p;$

18. Consider the following postfix expression.

1 2 3 4 + + 6 7 - - 9 0 * * 1 2 / /

If the expression is correctly formed, evaluate the expression and write the answer, otherwise answer No. [3]

19. If we assume that the address of i, p, and p1 are 1000, 2000, and 3000, respectively, what would be the output of the following program? [4]

```
int i=10, *p, **p1;

p = &i;
p1 = &p;

printf("%d %d %d ", i, *p, **p1);
printf("%p %p %p", &i, p, *p1);
```

20. What would be the output of the following program? [3]

```
#include <stdio.h>

void swap(char *p[])
{
    char *temp;

    temp = p[0];
    p[0] = p[1];
    p[1] = temp;
}

int main()
{
    char *p[3];

    p[0] = "Sachin";
    p[1] = "Ramesh";
    p[2] = "Tendulkar";

    swap(p);

    printf("%s %s %s", p[0], p[1], p[2]);
}
```

21. What would be the output of the following program? [3]

```
#include <stdio.h>

void fun(int (*b)[4])
{
    b[1][1] = 100;
    b++;
    b[0][1] = 10;
}

int main()
{
    int a[2][4] = {{1,2,3,4}, {5,6,7,8}};
```

```
fun(a);
printf("%d %d", a[0][1], a[1][1]);

return 0;
}
```

22. What would be the output of the following program? [3]

```
#include <stdio.h>

void fun(int *b)
{
    b[0] = 100;
}

int main(void)
{
    int a[] = { 1,2,3,4,5,6,7,8,9,10 };

    fun(a+5);

    printf("%d %d", a[0], a[5]);

    return 0;
}
```

23. What would be the output of the following program? [5]

```
#include <stdio.h>

int fun(int x, int y)
{
    if (x == 0)
        return y;
    else
        return fun(x - 1, x + y);
}

int main(void)
{
    printf("%d", fun(10, 2));
    return 0;
}
```

24. What would be the output of the following program? [4]

```
#include <stdio.h>

void find(int n)
{
    if (n == 0) return;
    printf("%d", n%2);
    find(n/2);
    printf("%d", n%2);
}
```

```
int main(void)
{
    find(12);
}
```

25. A queue is implemented using a non-circular singly linked list. The queue has a head/front pointer; additionally, it has a tail/rear pointer which points to the last node of the linked list. Let **Enqueue/Push** be implemented by inserting a new node at the head of the queue, and **Dequeue/Pop** be implemented by deleting a node from the tail. Consider the following two statements.

S1 : Enqueue/Push operation can be performed without traversing the list.

S2 : Dequeue/Pop operation can be performed without traversing the list.

Which of the above statements are true? [2]

- (a) Both **S1** and **S2** are FALSE.
- (b) **S1** is FALSE but **S2** is TRUE.
- (c) **S1** is TRUE but **S2** is FALSE.
- (d) Both **S1** and **S2** are TRUE.

26. Consider an empty stack **s**. You performed the following operations on the stack. State the values of **a**, **b**, and **c**. [2]

```
push(&s, 5)
push(&s, 6)
push(&s, 7)
```

```
a = pop(&s)
```

```
push(&s, 8)
```

```
b = pop(&s)
```

```
c = pop(&s)
```

27. Write the Postfix expression of the following Infix expression. [3]

```
((A - (B + C)) * D) * (E + (F - G))
```

28. Consider the following program?

```
#include<stdio.h>

void fun1(char *s1, char *s2)
{
    char *tmp;

    tmp = s1;
    s1 = s2;
    s2 = tmp;
}
```

```
void fun2(char **s1, char **s2)
{
    char *tmp;

    tmp = *s1;
    *s1 = *s2;
    *s2 = tmp;
}
```

```
int main(void)
{
    char *str1 = "IIT";
    char *str2 = "GHY";

    fun1(str1, str2);
    printf("%s %s ", str1, str2);

    fun2(&str1, &str2);
    printf("%s %s", str1, str2);

    return 0;
}
```

Which of the following is the output produced by it? . [4]

- (a) GHY IIT IIT GHY
- (b) IIT GHY GHY IIT
- (c) IIT GHY IIT GHY
- (d) GHY IIT GHY IIT

29. The following postfix expression with single digit operand is evaluated using stack. Note that **\$** represents the exponentiation operation. What are the top two elements of the stack after the first ***** is evaluated. [3]

```
8 2 3 $ / 2 3 * + 5 1 * -
```

30. Consider the following program for Bubble Sort.

```
#include <stdio.h>

void swap(int* xp, int* yp)
{
    int temp = *xp;

    *xp = *yp;
    *yp = temp;
}

void printArray(int arr[], int size)
{
    int i;

    for (i = 0; i < size; i++)
        printf("%d ", arr[i]);
}
```

```

    printf("\n");
}

void bubbleSort(int arr[], int n)
{
    int i, j;

    for (i = 0; i < n-1; i++)
    {
        for (j = 0; j < n-i-1; j++)
        {
            if (arr[j] > arr[j+1])
                swap(&arr[j], &arr[j+1]);

            if ( (i == 1) && (j == 1) )
                printArray(arr, n);
        }
    }
}

int main(void)
{
    int arr[] = {4, 34, 5, 1, 22, 10, 19};

    bubbleSort(arr, 7);

    return 0;
}

```

What would be the output of the program? [5]

- (a) 4 5 1 22 10 19 34
- (b) 4 1 5 22 10 19 34
- (c) 4 5 1 10 22 19 34
- (d) 4 5 1 22 10 34 19

31. A program P reads in 1000 integers in the range of [0...100] representing the score of 1000 students in **CS101** course. It then prints the frequencies of each score above 50. What would be the best way to store the frequencies? [3]

- (a) An array of 50 numbers.
- (b) An array of 100 numbers.
- (c) An array of 1000 numbers.
- (d) A dynamically allocated array of 1050 numbers.

32. Consider the binary search function, `binsrch`, given below. This function takes an array `arr` sorted in ascending order, the indices `l` and `r`, and a value `x`. It returns the location of `x` in the given array `arr[l..r]` if it is present, otherwise returns -1. Some of the expressions in the code is replaced by the letters A, B, C, and D.

```

int binsrch(int arr[], int l, int r, int x)
{
    int mid;

    if (l <= r)
    {
        mid = (l+r) / 2;

        if (arr[mid] == x)
            return mid;

        if (arr[mid] > x)
            return binsrch(arr, A, B, x);
        else
            return binsrch(arr, C, D, x);
    }
    else if (l > r)
    {
        return -1;
    }
}

```

Which of the following would be the correct expressions for these letters? [5]

- (a) A = mid+1, B = r, C = l, D = mid-1
- (b) A = l, B = mid+1, C = mid-1, D = r
- (c) A = mid, B = r, C = mid+1, D = l
- (d) A = l, B = mid-1, C = mid+1, D = r

33. Consider the following function to merge two sorted arrays. A is a sorted array of `m` elements and B is a sorted array of `n` elements. C is an array of size `m+n`.

```

void merge(int A[],int m,int B[],int n,int C[])
{
    int i = 0, j = 0, k = 0;

    while ( (i < m) && (j < n) )
    {
        if (A[i] < B[j])
            C[k++] = A[i++];
        else
            C[k++] = B[j++];
    }
}

```

If we call this function with `A[] = { 1, 6, 8, 10, 15, 20 }` and `B[] = { 1, 3, 7, 10, 11 }`, then which one of the following options is correct? [3]

- (a) Some elements of A won't be copied into C.
- (b) Some elements of B won't be copied into C.

- (c) Some elements from both A and B won't be copied into C.
- (d) All the elements of A and B will be copied to C.

34. Which of the following sorting algorithm needs an extra array to sort? [2]

- (a) Bubble Sort
- (b) Selection Sort
- (c) Insertion Sort
- (d) Merge Sort

35. Which of the following is a Last-in-First-out (LIFO) data structure? [2]

- (a) Stack
- (b) Queue
- (c) Both Stack and Queue
- (d) None of Stack and Queue

36. What would be the output of the following program? [4]

```
#include <stdio.h>

int func(int num)
{
    int count = 0;

    while (num)
    {
        count++;
        num >>= 1;
    }

    return count;
}

int main(void)
{
    printf("%d", func(435));

    return 0;
}
```

37. What would be the output of the following program? [3]

```
#include <stdio.h>

struct marks
{
    int s1;
    int s2;
    int s3;
};
```

```
int main(void)
{
    struct marks stud[3] = {{45, 68, 81},
                             {50, 60, 70}, {57, 91, 39}};

    printf("%d ", stud[1].s1);
    printf("%d ", stud[0].s2);
    printf("%d", stud[2].s3);

    return 0;
}
```

38. Consider the following two statements.

S1 : Insertion sort performs well when the array is almost sorted.

S2 : The number of swapping operations to be performed in Bubble sort is in general less than the number of swapping operations to be performed in Selection sort.

Which of the following statements is true? [2]

- (a) **S1** is TRUE and **S2** is FALSE.
- (b) **S1** is TRUE and **S2** is TRUE.
- (c) **S1** is FALSE and **S2** is FALSE.
- (d) **S1** is FALSE and **S2** is TRUE.

39. Consider the following program.

```
#include <stdio.h>

int main(void)
{
    int a = 10, b = 5;

    if(a && b)
        printf("Hello");
    else
        printf("Hi");

    return 0;
}
```

What would be its output? [2]

- (a) Hello
- (b) Hi

40. What would be the output of the following program? [3]

```
#include <stdio.h>

int main(void)
{
    int i, p;
```

```

    for (i=0; i<11; i++)
    {
        if (i == 0)
            p = 1;
        else
            p = p * 2;
    }

    printf("%d", p);

    return 0;
}

```

41. Consider the following piece of code.

```

int i, j, k;

for (i = 1; i <= 5; i++)
{
    for (j = 5; j >= 1; j--)
    {
        if (j == i)
            break;
    }
}

```

What is the value of j when both the loops have terminated? [3]

42. What will be printed by the following statement? [2]

```
printf("%5.3d", 10);
```

If you feel any blanks will be printed, you can represent them using \$ so that they will be visible to us.

43. What will be printed by the following statement? [2]

```
printf("%5.3f", 3.1412);
```

If you feel any blanks will be printed, you can represent them using \$ so that they will be visible to us.

44. What will be printed by the following statement? [2]

```
printf("%5.3s", "hello");
```

If you feel any blanks will be printed, you can represent them using \$ so that they will be visible to us.

45. Consider the following `scanf` statement.

```

int a, b;

scanf("%d%d", &a, &b);

```

On input

50t

what will be the values of a and b? [2]

- (a) Both a and b will have garbage values.
- (b) a will have the value 50 and b will have garbage value.
- (c) The program will not compile.
- (d) The program will compile but will give a runtime error.